

Building Your Data Science Portfolio

Jupyter Book 2 · MyST · GitHub Pages

A step-by-step guide to setting up, customizing, and deploying your portfolio — from extracting the ZIP to a live website.

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01 — Overview

This guide walks you through setting up, customizing, and deploying a professional Data Science portfolio built with Jupyter Book 2.

By the end, you will have:

- A local portfolio you can preview in your browser
- Jupyter notebooks styled as project case studies
- A live website at https://YOUR_USERNAME.github.io/ds-portfolio/
- A simple workflow for adding new projects throughout the semester

02 — What Changed from the Old Instructions

If your assignment mentions `_config.yml`, `_toc.yml`, `jupyter-book create`, or `ghp-import`, those are **Jupyter Book 1** commands. This portfolio uses **Jupyter Book 2**, the current version.

Old (JB 1)	Current (JB 2)
<code>_config.yml</code> + <code>_toc.yml</code>	<code>myst.yml</code> (single file)
<code>jupyter-book create</code>	<code>jupyter book init</code>
<code>jupyter-book build .</code>	<code>jupyter book build --html</code>
<code>ghp-import</code>	GitHub Actions (automatic)
Sphinx engine	MyST Document Engine

03 — Tools You Need

Python 3.10+

```
python --version
```

Download from python.org if not installed. On macOS/Linux you may need `python3` instead.

Git

```
git --version
```

Download from git-scm.com if not installed.

Node.js 18+

```
node --version
```

Jupyter Book 2 uses the MyST engine which requires Node.js. Download the LTS version from nodejs.org.

Code Editor

VS Code is recommended (code.visualstudio.com).

04 — Project Structure

```
ds-portfolio/  
  ■■■ myst.yml           Main configuration  
  ■■■ index.md           Homepage  
  ■■■ about.md           About page  
  ■■■ skills.md          Skills & toolkit  
  ■■■ README.md          Repository README  
  ■■■ requirements.txt    Python dependencies  
  ■■■ .gitignore         Git ignore rules  
  ■■■ assets/  
    ■ ■■■ custom.css     Portfolio styling  
  ■■■ projects/  
    ■ ■■■ index.md       Project gallery  
    ■ ■■■ lab*.ipynb     Lab notebooks  
  ■■■ .github/  
    ■■■ workflows/  
      ■■■ deploy.yml     Auto-deploy workflow
```

05 — Installing Jupyter Book

Step 1: Create a virtual environment

Windows PowerShell:

```
cd ds-portfolio  
python -m venv .venv  
.venv\Scripts\Activate.ps1
```

Windows Note

If PowerShell blocks the script, run this once:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

macOS / Linux:

```
cd ds-portfolio  
python3 -m venv .venv  
source .venv/bin/activate
```

Step 2: Install dependencies

```
python -m pip install --upgrade pip  
pip install -r requirements.txt
```

Step 3: Verify

```
jupyter book --version
```

You should see a version number starting with 2.x.

06 — Understanding the Files

myst.yml

The single configuration file. It defines project metadata, the table of contents (which pages appear and in what order), and site settings like the theme and custom CSS path.

index.md

Your homepage. Uses MyST Markdown — an extension of standard Markdown that supports directives like grid layouts and cards.

assets/custom.css

Controls visual appearance. The CSS variables at the top of the file define colors, spacing, and typography for the entire site. Change the accent color here to personalize the portfolio.

deploy.yml

The GitHub Actions workflow that automatically builds and publishes your site whenever you push to the main branch.

07 — Customizing Your Portfolio

Search for these placeholders across all files and replace them with your information:

Placeholder	Replace with
YOUR_NAME	Your full name
YOUR_PROGRAM	e.g., "BS Computer Science"
YOUR_YEAR_LEVEL	e.g., "3rd Year"
YOUR_UNIVERSITY	Your university
YOUR_GITHUB_USERNAME	Your GitHub username
YOUR_EMAIL	Your email
YOUR_LINKEDIN	Your LinkedIn URL slug
YOUR_DATE	Date of each lab

08 — Adding Notebooks

Each lab notebook contains structured placeholders marked with REPLACE comments. To fill in your actual work, open the notebook, replace placeholder cells, add your observations, run all cells, and save.

Adding a new notebook:

1. Create the .ipynb file in the projects/ directory
2. Add it to the toc section in myst.yml
3. Add a card entry to projects/index.md (optional)
4. Preview with jupyter book start
5. Commit and push to GitHub

TOC entry example:

```
# In myst.yml, under project.toc:  
- file: projects/my-new-notebook.ipynb
```

09 — Previewing Locally

```
jupyter book start
```

This starts a local development server. Open the URL shown (usually <http://localhost:3000>). Changes to .md files are reflected automatically. Press Ctrl+C to stop.

10 — Publishing to GitHub

Step 1: Create a repository

Go to github.com, click + then New Repository. Name it ds-portfolio. Choose Public. Do not initialize with README or .gitignore (you already have them).

Step 2: Push your code

```
git init  
git add .  
git commit -m "Initial Data Science portfolio"  
git branch -M main  
git remote add origin https://github.com/YOUR_USERNAME/ds-portfolio.git  
git push -u origin main
```

Important

Replace YOUR_USERNAME with your actual GitHub username.

11 — GitHub Pages Deployment

Enable GitHub Pages:

1. In your repository, go to Settings
2. Click Pages in the left sidebar
3. Under Source, select GitHub Actions
4. Push to main — the workflow handles the rest

Your site URL:

```
https://YOUR_USERNAME.github.io/ds-portfolio/
```

Tip

Check the Actions tab for build status.
Green checkmark = success. Red X = failure.
Click a failed run to see error details.

12 — Updating Your Portfolio

Every time you modify the portfolio, follow this routine:

```
git status
git add .
git commit -m "Add Lab 3 notebook"
git push
```

GitHub Actions automatically rebuilds and redeploys. You do not need `ghp-import` or manual build commands.

13 — Troubleshooting

jupyter: command not found

Your virtual environment is not activated. Run the activate command.

node: command not found

Node.js is not installed. Download from nodejs.org.

PowerShell execution policy error

Run: `Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser`

YAML indentation error

Use spaces not tabs. Each indent level is exactly 2 spaces.

Notebook missing from navigation

Every notebook must be listed in the toc section of `myst.yml`.

Broken project links

Card `:link:` value must match the filename without extension.

remote origin already exists

Run: `git remote remove origin`, then add origin again.

GitHub Pages shows 404

Verify Pages source is set to GitHub Actions. Wait 1-2 minutes.

GitHub Action failed

Check the Actions tab. Expand the failing step for error details.

CSS not loading

Verify the style path in `myst.yml` matches your CSS file location.

14 — Quick Command Reference

Start environment

Windows:

```
.venv\Scripts\Activate.ps1
```

macOS / Linux:

```
source .venv/bin/activate
```

Preview portfolio

```
jupyter book start
```

Build static HTML

```
jupyter book build --html
```

Update live site

```
git add .  
git commit -m "Update portfolio"  
git push
```

Add notebook to TOC

```
# In myst.yml under project.toc:  
- file: projects/your-notebook.ipynb
```

Generated for the Data Science Portfolio project. Keep this guide alongside your portfolio for reference.